DO NOW — RECALL

Lesson 108 · Interrupts and the Interrupt Service Routin

Name: _____ Class: _____ Date: _____

1. With page size 1,024 bytes, what offset is address 2,500?

2. For the binary value 1010, what is its decimal value?

3. A linked-list node has data and what other essential field?

4. In a BST, where is a value less than the current node placed?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

452.

2. Model answer

10.

3. Model answer

A link/reference to the next node (for a singly linked list).

4. Model answer

In its left subtree (under the chosen no-duplicate convention).

By the end, you can…

• A device or timer signals that attention is needed.

• The CPU completes a suitable instruction boundary and saves essential context.

• The ISR handles the event; saved context is restored and execution resumes.

Vocabulary — know these terms

interrupt — Signal requiring CPU attention.

ISR — Routine that services a particular interrupt.

context — CPU state needed to resume execution.

polling — Repeated checking of a device or condition.

NEW LEARNING

Read, annotate and explain

Event occurs

A device or timer signals that attention is needed.

State is preserved

The CPU completes a suitable instruction boundary and saves essential context.

Service and return

The ISR handles the event; saved context is restored and execution resumes.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A keyboard interrupt arrives while a calculation is running. Trace the CPU response.

Model answer

After completing the current instruction and recognising the interrupt, the CPU preserves relevant state, runs the keyboard ISR to obtain the key event, restores state and resumes the interrupted program.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Outline the role of an interrupt and an ISR. [6]

Question 2

Explain where an interrupt can fit into the fetch–decode–execute cycle. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate interrupts compared with repeatedly polling a low-frequency sensor. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An ISR should normally preserve enough execution state to allow the interrupted work to resume.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.